

Quiz 12.5

Estimate the Gaussian (0,1) random variable X using the observation vector $\mathbf{Y} = \mathbf{1}X + \mathbf{W}$, where $\mathbf{1}$ is the vector of 20 1's. The noise vector $\mathbf{W} = [W_0 \ \cdots \ W_{19}]'$ is independent of X , has zero expected value, and has a correlation matrix with i, j th entry $R_{\mathbf{W}}(i, j) = c^{|i-j|-1}$. Find $\hat{X}_L(\mathbf{Y})$, the linear MMSE estimate of X given $\mathbf{Y}$. For c in the range $0 < c < 1$, what value of c minimizes the mean square error of the estimate?

Further Reading: The final chapter of [WS01] presents the basic theory of estimation of random variables as well as extensions to stochastic process estimation in the time domain and frequency domain.

Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

12.1.1 ● Generalizing the solution of Example 12.2, let the call duration T be an exponential (λ) random variable. For $t_0 > 0$, show that the minimum mean square error estimate of T , given that $T > t_0$ is

$$\hat{T} = t_0 + E[T].$$

12.1.2 ● X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6(y-x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_X(x)$?
- (b) What is the blind estimate $\hat{x}_B$?
- (c) What is the minimum mean square error estimate of X given $X < 0.5$?
- (d) What is $f_Y(y)$?
- (e) What is the blind estimate $\hat{y}_B$?
- (f) What is the minimum mean square error estimate of Y given $Y > 0.5$?

12.1.3 ■ X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_X(x)$?
- (b) What is the blind estimate $\hat{x}_B$?

- (c) What is the minimum mean square error estimate of X given $X > 1/2$?
- (d) What is $f_Y(y)$?
- (e) What is the blind estimate $\hat{y}_B$?
- (f) What is the minimum mean square error estimate of Y given $X > 1/2$?

12.1.4 ● X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 6(y-x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is $f_{Y|X}(y|x)$?
- (d) What is $\hat{y}_M(x)$, the MMSE estimate of Y given $X = x$?

12.1.5 ■ X and Y have the joint PDF

$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is

$$e^*(0.5) = E[(X - \hat{x}_M(0.5))^2 | Y = 0.5],$$

the minimum mean square error of the estimate of X given $Y = 0.5$?

12.1.6 ■ A signal X and noise Z are independent Gaussian $(0, 1)$ random variables, and $Y = X + Z$ is a noisy observation of the signal X . Usually, we want to use Y to estimate of X ; however, in this problem we will use Y to estimate the noise Z .

- Find $\hat{Z}(Y)$, the MMSE estimator of Z given Y .
- Find the mean squared error $e = E[(Z - \hat{Z}(Y))^2]$.

12.1.7 ♦ Random variable $Y = X - Z$ is a noisy observation of the continuous random variable X . The noise Z has zero expected value and unit variance and is independent of X . Find the conditional expectation $E[X|Y]$.

12.1.8 ♦ In a BPSK communication system, a source wishes to communicate a random bit X to a receiver. The possible inputs $X = 1$ and $X = -1$ are equally likely. In this system, the source transmits X multiple times. In the i th transmission, the receiver observes $Y_i = X + W_i$. After n transmissions of X , the receiver has observed $\mathbf{Y} = \mathbf{y} = [y_1 \ \cdots \ y_n]'$.

- Find $\hat{X}_n(\mathbf{y})$, the MMSE estimate of X given the observation $\mathbf{Y} = \mathbf{y}$. Express your answer in terms of the likelihood ratio

$$L(\mathbf{y}) = \frac{f_{\mathbf{Y}|X}(\mathbf{y}|1)}{f_{\mathbf{Y}|X}(\mathbf{y}|-1)}.$$

- Simplify your answer when the W_i are iid Gaussian $(0, 1)$ random variables, independent of X .

12.2.1 ■ Random variables X and Y have joint PMF

$P_{X,Y}(x, y)$	$y = -3$	$y = -1$	$y = 1$	$y = 3$
$x = -1$	1/6	1/8	1/24	0
$x = 0$	1/12	1/12	1/12	1/12
$x = 1$	0	1/24	1/8	1/6

- Find the marginal probability mass functions $P_X(x)$ and $P_Y(y)$.
- Are X and Y independent?
- Find $E[X]$, $\text{Var}[X]$, $E[Y]$, $\text{Var}[Y]$, and $\text{Cov}[X, Y]$.
- Let $\hat{X}(Y) = aY + b$ be a linear estimator of X . Find a^* and b^* , the values of a and b that minimize the mean square error e_L .
- What is e_L^* , the minimum mean square error of the optimum linear estimate?
- Find $P_{X|Y}(x|-3)$, the conditional PMF of X given $Y = -3$.
- Find $\hat{x}_M(-3)$, the optimum (nonlinear) mean square estimator of X given $Y = -3$.
- Find the mean square error

$$e^*(-3) = E[(X - \hat{x}_M(-3))^2 | Y = -3]$$

of this estimate.

12.2.2 ● A telemetry voltage V , transmitted from a position sensor on a ship's rudder, is a random variable with PDF

$$f_V(v) = \begin{cases} 1/12 & -6 \leq v \leq 6, \\ 0 & \text{otherwise.} \end{cases}$$

A receiver in the ship's control room receives $R = V + X$. The random variable X is a Gaussian $(0, \sqrt{3})$ noise voltage that is independent of V . The receiver uses R to calculate a linear estimate of the telemetry voltage: $\hat{V} = aR + b$. Find

- the expected received voltage $E[R]$,
- the variance $\text{Var}[R]$ of the received voltage,
- the covariance $\text{Cov}[V, R]$ of the transmitted and received voltages,
- a^* and b^* , the optimum coefficients in the linear estimate,
- e_L^* , the minimum mean square error of the estimate.

12.2.3 Random variables X and Y have joint PMF given by the following table:

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$
$x = -1$	3/16	1/16	0
$x = 0$	1/6	1/6	1/6
$x = 1$	0	1/8	1/8

We estimate Y by $\hat{Y}_L(X) = aX + b$.

- Find a and b to minimize the mean square estimation error.
- What is the minimum mean square error e_L^* ?

12.2.4 The random variables X and Y have the joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} 2(y+x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

What is $\hat{X}_L(Y)$, the linear minimum mean square error estimate of X given Y ?

12.2.5 For random variables X and Y from Problem 12.1.4, find $\hat{X}_L(Y)$, the linear minimum mean square error estimator of X given Y .

12.2.6 Random variable X has a second-order Erlang PDF

$$f_X(x) = \begin{cases} \lambda x e^{-\lambda x} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Given $X = x$, Y is a uniform $(0, x)$ random variable. Find

- the MMSE estimate of Y given $X = x$, $\hat{y}_M(x)$,
- the MMSE estimate of X given $Y = y$, $\hat{x}_M(y)$,
- the LMSE estimate of Y given X , $\hat{Y}_L(X)$,
- the LMSE estimate of X given Y , $\hat{X}_L(Y)$.

12.2.7 Random variable R has an exponential PDF with expected value 1. Given $R = r$, X has an exponential PDF with expected value $1/r$. Find

- the MMSE estimate of R given $X = x$, $\hat{r}_M(x)$,
- the MMSE estimate of X given $R = r$, $\hat{x}_M(r)$,
- the LMSE estimate of R given X , $\hat{R}_L(X)$,
- the LMSE estimate of X given R , $\hat{X}_L(R)$.

12.2.8 For random variables X and Y , we wish to use Y to estimate X . However, our estimate must be of the form $\hat{X} = aY$.

- Find a^* , the value of a that minimizes the mean square error $e = E[(X - aY)^2]$.
- For $a = a^*$, what is the minimum mean square error e^* ?
- Under what conditions is $\hat{X}$ the LMSE estimate of X ?

12.2.9 Here are four different joint PMFs:

$P_{X,Y}(x,y)$	$x = -1$	$x = 0$	$x = 1$
$y = -1$	1/9	1/9	1/9
$y = 0$	1/9	1/9	1/9
$y = 1$	1/9	1/9	1/9

$P_{U,V}(u,v)$	$u = -1$	$u = 0$	$u = 1$
$v = -1$	0	0	1/3
$v = 0$	0	1/3	0
$v = 1$	1/3	0	0

$P_{S,T}(s,t)$	$s = -1$	$s = 0$	$s = 1$
$t = -1$	1/6	0	1/6
$t = 0$	0	1/3	0
$t = 1$	1/6	0	1/6

$P_{Q,R}(q,r)$	$q = -1$	$q = 0$	$q = 1$
$r = -1$	1/12	1/12	1/6
$r = 0$	1/12	1/6	1/12
$r = 1$	1/6	1/12	1/12

- For each pair of random variables, indicate whether the two random variables are independent, and compute the correlation coefficient ρ .

- (b) Compute the least mean square linear estimator $\hat{U}_L(V)$ of U given V . What is the mean square error? Do the same for the pairs X, Y , Q, R , and S, T .

12.2.10 ♦ Random variable $Y = X - Z$ is a noisy observation of the continuous random variable X . The noise Z has zero expected value and unit variance and is independent of X . Consider the following argument:

Since $X = Y + Z$, we see that if $Y = y$, then $X = y + Z$. Thus, by Theorem 6.4, the conditional PDF of X given $Y = y$ is $f_{X|Y}(x|y) = f_Z(x - y)$. It follows that

$$\begin{aligned} E[X|Y = y] &= \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx \\ &= \int_{-\infty}^{\infty} x f_Z(x - y) dx. \end{aligned}$$

With the variable substitution, $z = x - y$,

$$\begin{aligned} E[X|Y = y] &= \int_{-\infty}^{\infty} (z + y) f_Z(z) dz \\ &= E[Z] + y = y. \end{aligned}$$

We conclude that $E[X|Y] = Y$. Since $E[X|Y]$ is optimal in the mean square sense, we conclude that the optimal linear estimator $\hat{X}(Y) = \hat{a}Y$ must satisfy $\hat{a} = 1$.

Prove that this conclusion is wrong. What is the error in the above argument? Hint: Find the LMSE estimator $\hat{X}_L(Y) = aY$.

12.3.1 ■ Suppose that in Quiz 12.3, R , measured in meters, has a uniform PDF over $[0, 1000]$. Find the MAP estimate of R given $X = x$. In this case, are the MAP and ML estimators the same?

12.3.2 ♦ Let R be an exponential random variable with expected value $1/\mu$. If $R = r$, then over an interval of length T , the number of phone calls N that arrive at a telephone switch has a Poisson PMF with expected value rT .

- (a) Find the MMSE estimate of N given R .
 (b) Find the MAP estimate of N given R .
 (c) Find the ML estimate of N given R .

12.3.3 ♦ Let R be an exponential random variable with expected value $1/\mu$. If $R = r$, then over an interval of length T , the number of phone calls N that arrive at a telephone switch has a Poisson PMF with expected value rT .

- (a) Find the MMSE estimate of R given N .
 (b) Find the MAP estimate of R given N .
 (c) Find the ML estimate of R given N .

12.3.4 ♦ Flip a coin n times. For each flip, the probability of heads is $Q = q$ independent of all other flips. Q is a uniform $(0, 1)$ random variable. K is the number of heads in n flips.

- (a) What is the ML estimator of Q given K ?
 (b) What is the PMF of K ? What is $E[K]$?
 (c) What is the conditional PDF $f_{Q|K}(q|k)$?
 (d) Find the MMSE estimator of Q given $K = k$.

12.4.1 ● You would like to know a sample value of X , a Gaussian $(0, 4)$ random variable. However, you only can observe noisy observations of the form $Y_i = X + N_i$. In terms of a vector of noisy observations, you observe

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} X + \begin{bmatrix} N_1 \\ N_2 \end{bmatrix},$$

where N_1 is a Gaussian $(0, 1)$ random variable and N_2 is a Gaussian $(0, 2)$ random variable. Under the assumption that X , N_1 , and N_2 are mutually independent, answer the following questions:

- (a) Suppose you use Y_1 as an estimate of X . The error in the estimate is $D_1 = Y_1 - X$. What are the expected error $E[D_1]$ and the expected squared error $E[D_1^2]$?
 (b) Suppose we use $Y_3 = (Y_1 + Y_2)/2$ as an estimate of X . The error for this estimate is $D_3 = Y_3 - X$. Find the expected squared error $E[D_3^2]$. Is Y_3 or Y_1 a better estimate of X ?

- (c) Let $Y_4 = \mathbf{A}\mathbf{Y}$ where $\mathbf{A} = \begin{bmatrix} a & 1-a \end{bmatrix}$ is a 1×2 matrix. Let $D_4 = Y_4 - X$ denote the error in using Y_4 as an estimate for X . In terms of a , what is the expected squared error $E[D_4^2]$? What value of a minimizes $E[D_4^2]$?

12.4.2 $\bullet$ $\mathbf{X}$ is a three-dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and autocorrelation matrix $\mathbf{R}_{\mathbf{X}}$ with elements $r_{ij} = (-0.80)^{|i-j|}$. Use X_1 and X_2 to form a linear estimate of X_3 : $\hat{X}_3 = a_1X_2 + a_2X_1$.

- (a) What are the optimum coefficients $\hat{a}_1$ and $\hat{a}_2$ and corresponding minimum mean square error e_L^* ?
- (b) Use X_2 to form a linear estimate of X_3 : $\hat{X}_3 = aX_2 + b$. What are the optimum coefficients a^* and b^* and corresponding minimum mean square error e_L^* ?

12.4.3 $\blacksquare$ $\mathbf{X}$ is a 3-dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and autocorrelation matrix $\mathbf{R}_{\mathbf{X}}$ with elements

$$R_{\mathbf{X}}(i, j) = 1 - 0.25|i - j|.$$

$\mathbf{Y}$ is a two-dimensional random vector with

$$Y_1 = X_1 + X_2, \quad Y_2 = X_2 + X_3.$$

Use $\mathbf{Y}$ to form $\hat{X}_1 = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \mathbf{Y}$, a linear estimate of X_1 .

- (a) Find the optimum coefficients $\hat{a}_1$ and $\hat{a}_2$ and the minimum mean square error e_L^* .
- (b) Use Y_1 to form a linear estimate of X_1 : $\hat{X}_1 = aY_1 + b$. What are the optimum coefficients a^* and b^* ? What is the minimum mean square error e_L^* ?

12.4.4 $\blacksquare$ $\mathbf{X}$ is a three-dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and correlation matrix $\mathbf{R}_{\mathbf{X}}$ with elements

$$R_{\mathbf{X}}(i, j) = 1 - 0.25|i - j|.$$

$\mathbf{W}$ is a two-dimensional random vector, independent of $\mathbf{X}$, with $E[\mathbf{W}] = \mathbf{0}$, $E[W_1W_2] = 0$, and $E[W_1^2] = E[W_2^2] = 0.1$.

$\mathbf{Y}$ is a two-dimensional random vector with

$$\begin{aligned} Y_1 &= X_1 + X_2 + W_1, \\ Y_2 &= X_2 + X_3 + W_2. \end{aligned}$$

Use $\mathbf{Y}$ to form $\hat{X}_1 = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \mathbf{Y}$, a linear estimate of X_1 .

- (a) Find the optimum coefficients $\hat{a}_1$ and $\hat{a}_2$ and minimum mean square error e_L^* .
- (b) Use Y_1 to form a linear estimate of X_1 : $\hat{X}_1 = aY_1 + b$. What are the optimum coefficients a^* and b^* ? What is the minimum mean square error e_L^* ?

12.4.5 $\blacksquare$ Suppose

$$Y_k = q_0 + q_1k + q_2k^2 + Z_k,$$

where $q_0 + q_1k + q_2k^2$ is an unknown quadratic function of k and Z_k is a sequence of iid Gaussian $(0, 1)$ noise random variables. We wish to estimate the unknown parameters q_0 , q_1 , and q_2 of the quadratic function. Suppose we assume q_0 , q_1 , and q_2 are samples of iid Gaussian $(0, 1)$ random variables. Find the optimum linear estimator $\hat{\mathbf{Q}}(\mathbf{Y})$ of $\mathbf{Q} = [q_0 \ q_1 \ q_2]'$ given the observation $\mathbf{Y} = [Y_1 \ \cdots \ Y_n]'$.

12.4.6 $\blacksquare$ $\mathbf{X}$ is a three-dimensional random vector with $E[\mathbf{X}] = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}'$ and correlation matrix $\mathbf{R}_{\mathbf{X}}$ with elements

$$R_{\mathbf{X}}(i, j) = 1 - 0.25|i - j|.$$

$\mathbf{W}$ is a two-dimensional random vector, independent of $\mathbf{X}$, with $E[\mathbf{W}] = \mathbf{0}$, $E[W_1W_2] = 0$, and

$$E[W_1^2] = E[W_2^2] = 0.1.$$

$\mathbf{Y}$ is a two-dimensional random vector with

$$\begin{aligned} Y_1 &= X_1 + X_2 + W_1, \\ Y_2 &= X_2 + X_3 + W_2. \end{aligned}$$

Use $\mathbf{Y}$ to form a linear estimate of X_1 :

$$\hat{X}_1 = \begin{bmatrix} \hat{a}_1 & \hat{a}_2 \end{bmatrix} \mathbf{Y} + \hat{b}.$$

- (a) What are the optimum coefficients $\hat{a}_1$, $\hat{a}_2$, and $\hat{b}$?

- (b) Find the MMSE e_L^* .
- (c) Use Y_1 to form a linear estimate of X_1 : $\hat{X}_1 = aY_1 + b$. What are the optimum coefficients a^* and b^* ? What is the minimum mean square error e_L^* ?

12.4.7 ■ When X and Y have expected values $\mu_X = \mu_Y = 0$, Theorem 12.3 says that $\hat{X}_L(Y) = \rho_{X,Y} \frac{\sigma_X}{\sigma_Y} Y$. Show that this result is a special case of Theorem 12.8 when random vector $\mathbf{Y}$ is the one-dimensional random variable Y .

12.4.8 ■ Prove the following theorem: $\mathbf{X}$ is an n -dimensional random vector with $E[\mathbf{X}] = \mathbf{0}$ and autocorrelation matrix $\mathbf{R}_X$ with elements $r_{ij} = c^{|i-j|}$, where $|c| < 1$. The optimum linear estimator of X_n ,

$$\hat{X}_n = a_1 X_{n-1} + a_2 X_{n-2} + \cdots + a_{n-1} X_1,$$

is $\hat{X}_n = cX_{n-1}$. The minimum mean square estimation error is $e_L^* = 1 - c^2$. Hint: Consider the $n-1$ equations $\partial e_L / \partial a_i = 0$.

12.4.9 ♦ In the CDMA multiuser communications system introduced in Problem 11.3.8, each user i transmits an independent data bit X_i such that the vector $\mathbf{X} = [X_1 \cdots X_n]'$ has iid components with $P_{X_i}(1) = P_{X_i}(-1) = 1/2$. The received signal is

$$\mathbf{Y} = \sum_{i=1}^k X_i \sqrt{p_i} \mathbf{S}_i + \mathbf{N},$$

where $\mathbf{N}$ is a Gaussian $(\mathbf{0}, \sigma^2 \mathbf{I})$ noise.

- (a) Based on the observation $\mathbf{Y}$, find the LMSE estimate $\hat{X}_i(\mathbf{Y}) = \hat{\mathbf{a}}_i' \mathbf{Y}$ of X_i .
- (b) Let $\hat{\mathbf{X}} = [\hat{X}_1 \cdots \hat{X}_k]'$ denote the vector of LMSE estimates of bits transmitted by users $1, \dots, k$. Show that

$$\hat{\mathbf{X}} = \mathbf{P}^{1/2} \mathbf{S}' (\mathbf{S} \mathbf{P} \mathbf{S}' + \sigma^2 \mathbf{I})^{-1} \mathbf{Y}.$$

12.5.1 ● Continuing Example 12.10, the 21-dimensional vector $\mathbf{X}$ has correlation ma-

trix $\mathbf{R}_X$ with i, j th element

$$R_X(i, j) = \frac{\sin(\phi_0 \pi |i - j|)}{\phi_0 \pi |i - j|}.$$

We use the observation vector $\mathbf{Y} = \mathbf{Y}_{(n)} = [Y_1 \cdots Y_n]'$ to estimate $X = X_{21}$. Find the LMSE estimate $\hat{X}_L(\mathbf{Y}_{(n)}) = \hat{\mathbf{a}}_{(n)}' \mathbf{Y}_{(n)}$. Graph the mean square error $e_L^*(n)$ as a function of the number of observations n for $\phi_0 \in \{0.1, 0.5, 0.9\}$. Interpret your results. Does smaller ϕ_0 or larger ϕ_0 yield better estimates?

12.5.2 ● Repeat Problem 12.5.1 when

$$R_X(i, j) = \cos(\phi_0 \pi |i - j|).$$

12.5.3 ■ In a variation on Example 12.10, we use the observation vector $\mathbf{Y} = \mathbf{Y}_{(n)} = [Y_1 \cdots Y_n]'$ to estimate $X = X_1$. The 21-dimensional vector $\mathbf{X}$ has correlation matrix $\mathbf{R}_X$ with i, j th element

$$R_X(i, j) = r_{|i-j|}.$$

Find the LMSE estimate $\hat{X}_L(\mathbf{Y}_{(n)}) = \hat{\mathbf{a}}_{(n)}' \mathbf{Y}_{(n)}$. Graph the mean square error $e_L^*(n)$ as a function of the number of observations n , and interpret your results for the cases

- (a) $r_{|i-j|} = \frac{\sin(0.1\pi|i-j|)}{0.1\pi|i-j|}$,
- (b) $r_{|i-j|} = \cos(0.5\pi|i-j|)$.

12.5.4 ♦♦ In the k user CDMA system employing LMSE receivers in Problem 12.4.9, the receiver employs the LMSE bit estimate $\hat{X}_i$ to implement the bit decision rule $\hat{X}_i = \text{sgn}(\hat{X}_i)$ for user i . Using the approach in Problem 11.4.6, construct a simulation to estimate the BER for a system with processing gain $n = 32$, with each user operating at 6 dB SNR. Graph your results as a function of k for $k = 1, 2, 4, 8, 16, 32$. Make sure to average your results over the choice of code vectors $\mathbf{S}_i$.